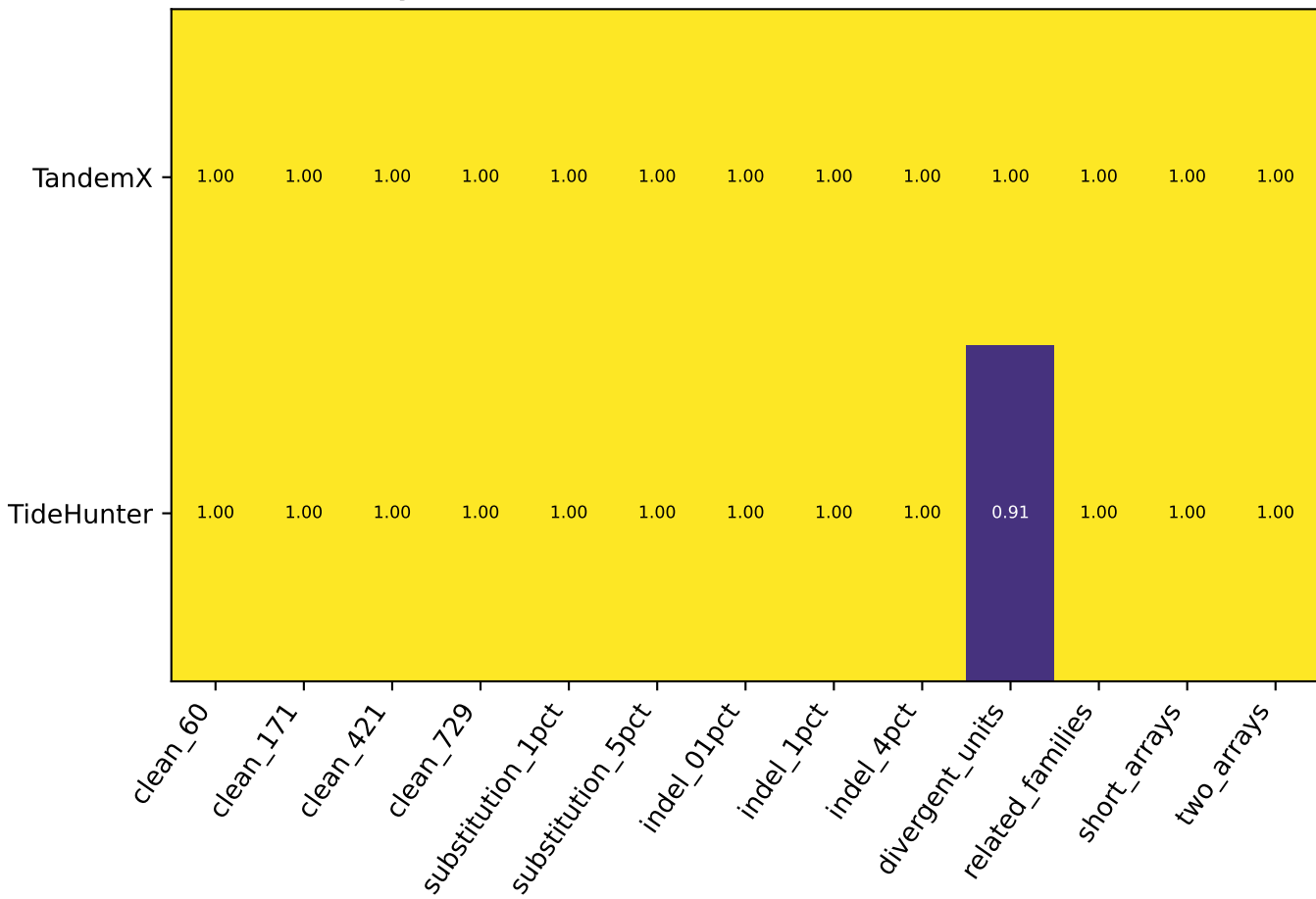
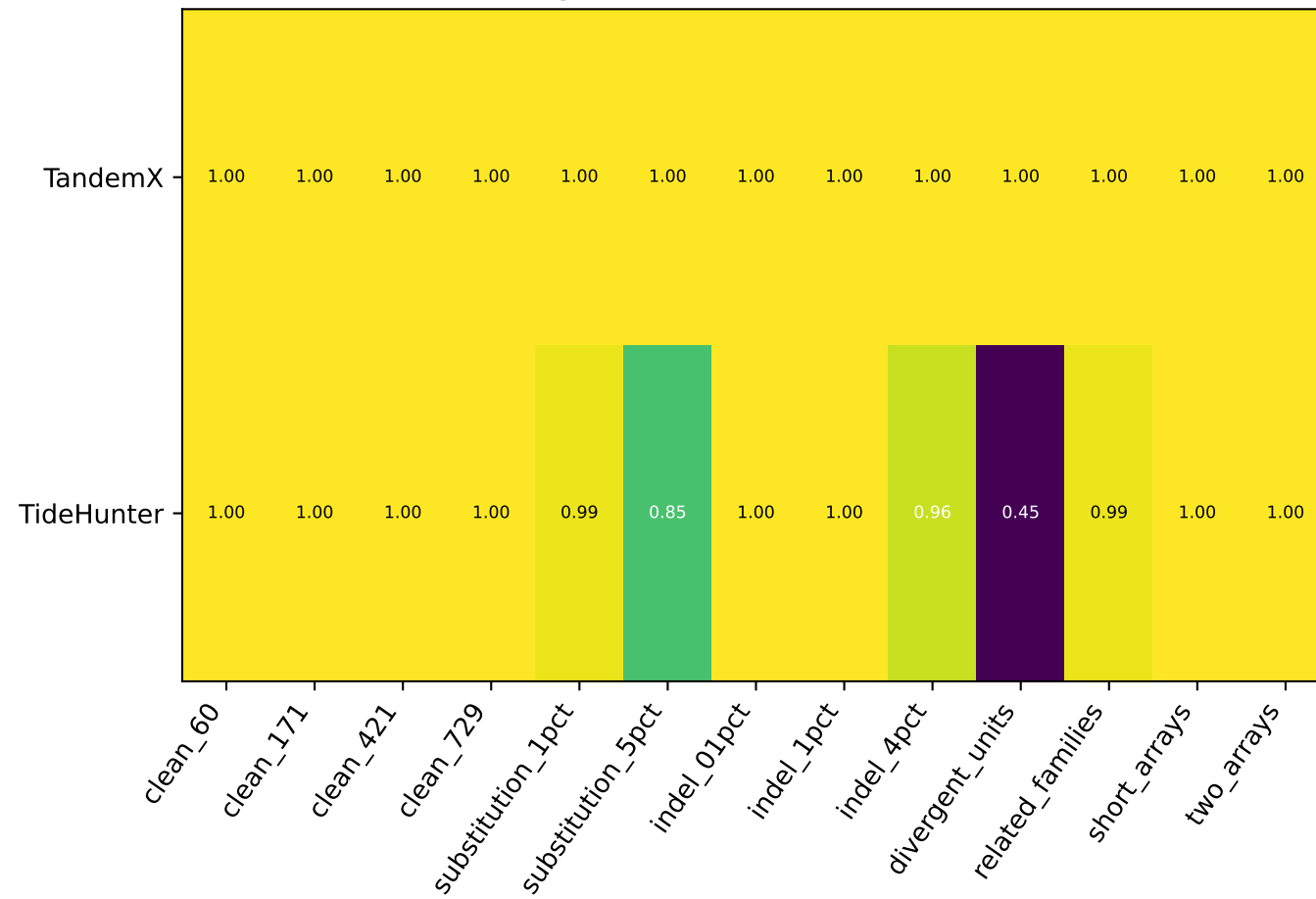


Cascade discovery validation audit: all 14 predeclared gates pass

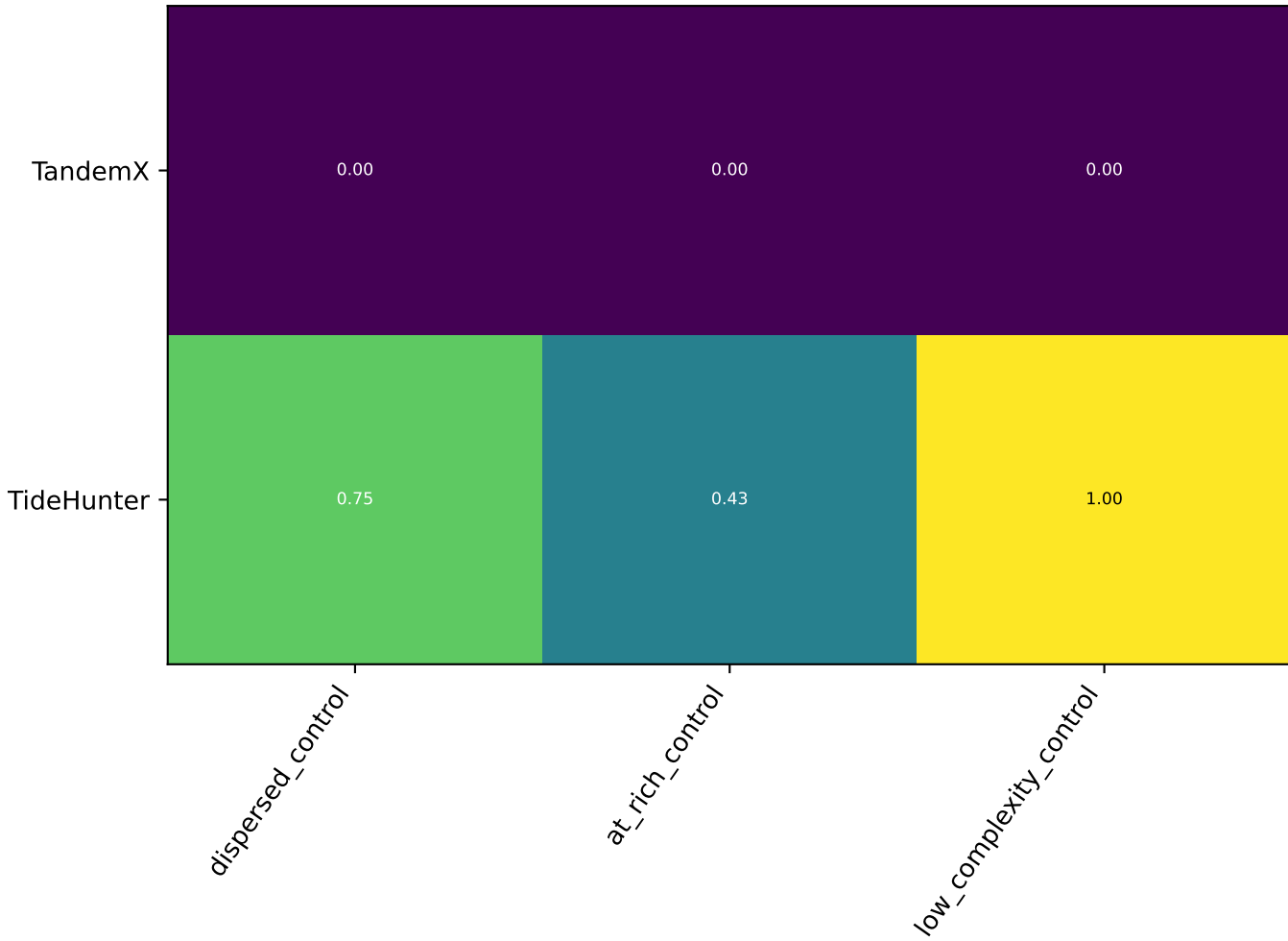
A Mean array recall across validation seeds



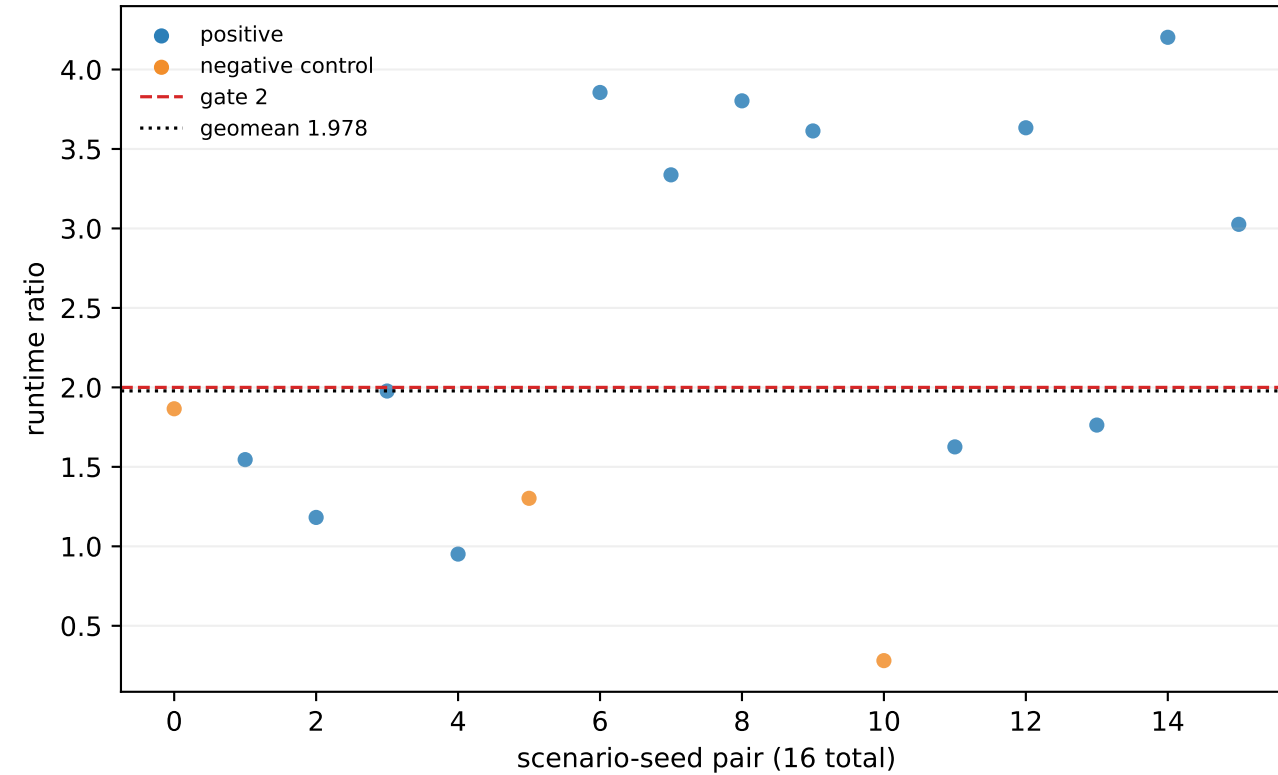
B Mean raw-call array precision



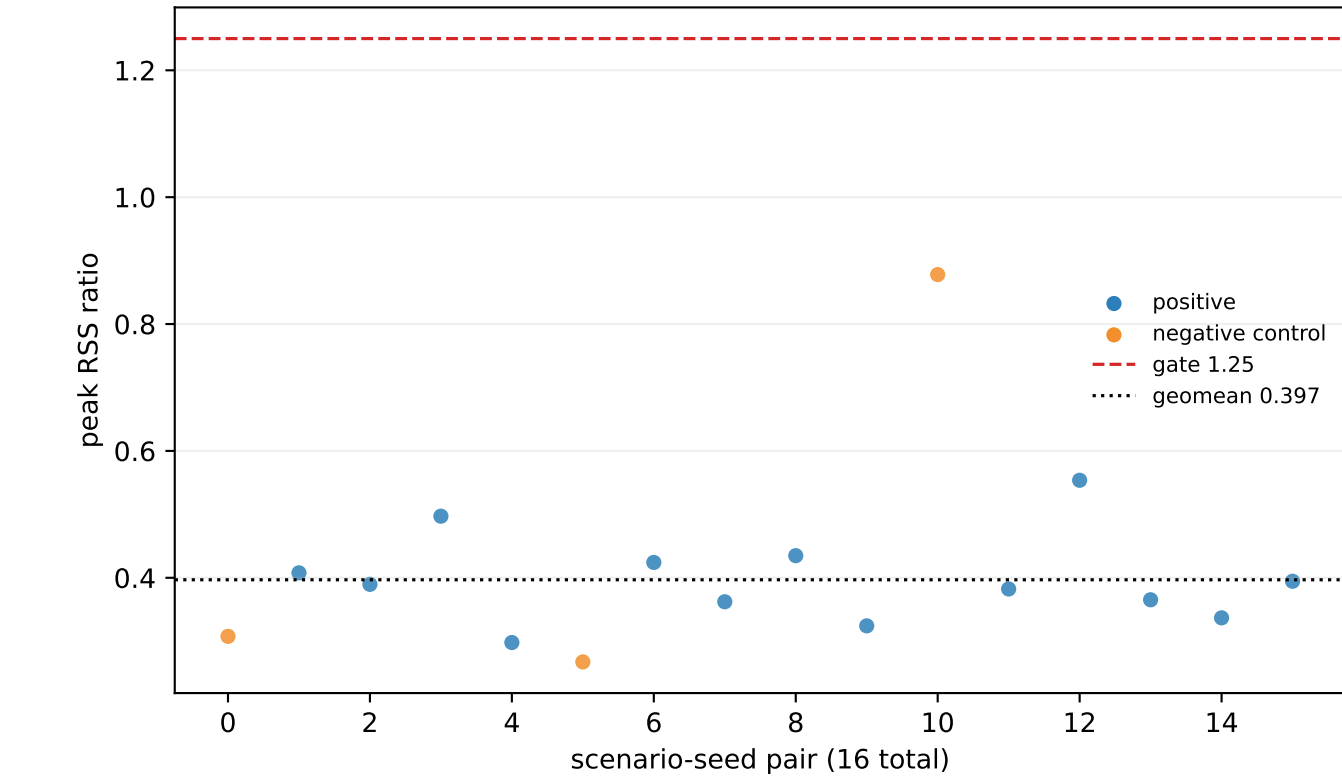
C Negative-read call rate (NA = process failure)



D Paired TandemX/TideHunter wall-time ratio



E Paired TandemX/TideHunter peak-RSS ratio



F Predeclared evaluation gates

